

AP Statistics MCQ Practice Exam 08

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AP Statistics MCQ Practice Exam 8

PART A, QUESTIONS

APSTAT-MCQ8-001

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A nonnegative integer random variable X can equal 0, 1, 2, or 3. Use the identity $E(X) = P(X \geq 1) + P(X \geq 2) + P(X \geq 3)$. If these three probabilities are 0.80, 0.50, and 0.20, respectively, what is $E(X)$?

- A) 0.80
- B) 1.30
- C) 1.50
- D) 2.30

APSTAT-MCQ8-002

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

In a boxplot, the median is close to Q_1 , and the upper half of the box and upper whisker are much longer than their lower counterparts. Which statement is most appropriate?

- A) The display suggests right skew, although a boxplot does not show every feature of the distribution.
- B) The distribution must be exactly normal because the median lies inside the box.
- C) The mean must equal Q_3 because the upper whisker is long.
- D) The display proves that more than 75% of observations are above the median.

APSTAT-MCQ8-003

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Two two-sided t-tests have the same value $|t| = 2.1$. Test 1 has $df = 5$, and test 2 has $df = 50$. Which statement is correct about their p-values?

- A) Test 1 has the smaller p-value because fewer observations create stronger evidence.
- B) The p-values are equal because the t statistics have equal magnitudes.
- C) Test 2 has p-value 0 because df is large.
- D) Test 2 has the smaller p-value because the t distribution has lighter tails at the larger degrees of freedom.

APSTAT-MCQ8-004

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A company compares independent random samples processed by an automated method and a manual method. It seeks evidence that the automated method has a higher population on-time proportion. Which hypotheses are appropriate?

- A) $H_0: p_{\text{auto}} > p_{\text{manual}}$; $H_a: p_{\text{auto}} = p_{\text{manual}}$
- B) $H_0: p_{\text{auto}} = p_{\text{manual}}$; $H_a: p_{\text{auto}} > p_{\text{manual}}$
- C) $H_0: \hat{p}_{\text{auto}} = \hat{p}_{\text{manual}}$; $H_a: \hat{p}_{\text{auto}} > \hat{p}_{\text{manual}}$
- D) $H_0: p_{\text{auto}} = p_{\text{manual}}$; $H_a: p_{\text{auto}} \neq p_{\text{manual}}$

APSTAT-MCQ8-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A regression model predicts equipment energy use y from operating load x . At $x = 8$, the predicted energy use is 26 units, and the least-squares slope is 2.5. What is the intercept of the regression line?

- A) 46
- B) 20
- C) 6
- D) 3.25

APSTAT-MCQ8-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

In an experiment comparing two visible physical-therapy programs, participants and therapists cannot be blinded. Which step would still help reduce outcome-assessment bias?

- A) Let each therapist score only participants who received that therapist's preferred program.
- B) Tell assessors which program is expected to work better.
- C) Allow participants to choose which assessor evaluates them.
- D) Have assessors who do not know treatment assignments evaluate the final mobility recordings.

APSTAT-MCQ8-007

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A manufacturer wants to estimate the probability that an item fails an initial inspection while avoiding strong dependence among observations from the same production batch. Which plan is best?

- A) Randomly select many batches and inspect one randomly selected item from each selected batch.
- B) Inspect every item from one conveniently selected batch.
- C) Inspect the first item produced after each supervisor's lunch break.
- D) Ask supervisors to nominate items that appear likely to fail.

APSTAT-MCQ8-008

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A sample proportion is $\hat{p} = 0.32$, and its estimated standard error for a confidence interval is 0.020. Approximately what sample size produced this standard error?

- A) 272
- B) 544
- C) 800
- D) 1,088

APSTAT-MCQ8-009

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Dataset A has standard deviation 2 meters. Dataset B measures the same type of quantity and has standard deviation 150 centimeters. Which dataset has greater spread as measured by standard deviation?

- A) Dataset B, because 150 is greater than 2
- B) The spreads are equal because both standard deviations are positive
- C) The comparison is impossible unless the means are equal
- D) Dataset A, because 2 meters equals 200 centimeters

APSTAT-MCQ8-010

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

Employees are selected by a simple random sample from a company and then randomly assigned to one of two break schedules. A confidence interval for the mean treatment effect on completed tasks is entirely positive. Which scope statement is justified?

- A) The result is causal but applies only to the particular employees observed, despite the random sample.
- B) The result generalizes to all workers everywhere because assignment was random.
- C) The design supports a causal conclusion for, and generalization to, the company's employee population represented by the sampling frame.
- D) The design supports association only because confidence intervals cannot be used with experiments.

APSTAT-MCQ8-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A study assigns participants different numerical exercise durations and measures resting heart rate before and after the program. The goal is to model how heart-rate change is associated with assigned

exercise duration. Which formulation is appropriate?

- A) For each participant, use assigned exercise duration as x and after-minus-before heart-rate change as y .
- B) Use participant identity as x and assigned duration as y .
- C) Use one overall mean duration as x and one overall mean heart rate as y .
- D) Use before-program heart rate as both x and y .

APSTAT-MCQ8-012

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Two mutually exclusive causes account for an alarm. The probability of cause A is 0.30 and the probability of cause B is 0.20. Given that the alarm was caused by either A or B, what is the probability that A was the cause?

- A) 0.50
- B) 0.60
- C) 0.30
- D) 0.10

APSTAT-MCQ8-013

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A retailer wants to estimate the proportion of item-location-time combinations for which a product is out of stock. Which plan best defines a probability sample from the target population?

- A) Check products that customers report as unavailable.
- B) Inspect only the highest-selling product at noon each day.
- C) Randomly select combinations from a complete list of products, store locations, and eligible time blocks, then record stock status.
- D) Ask store managers to choose combinations that seem typical.

APSTAT-MCQ8-014

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A quantitative dataset has sample variance 4. A transformed variable is defined by $y = -3x + 10$. What is the sample variance of y ?

- A) -12
- B) 4
- C) 12
- D) 36

APSTAT-MCQ8-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

For a random sample of days, a forecast error is defined as actual high temperature minus forecast high temperature. Which procedure addresses whether the forecasting method is unbiased on average?

- A) A two-proportion z-test comparing warm and cool days
- B) A one-sample t-test of $H_0: \mu_{\text{error}} = 0$ against $H_a: \mu_{\text{error}} \neq 0$
- C) A paired t-test pairing unrelated days from different months
- D) A two-sample t-test treating actual and forecast temperatures as independent groups

Questions APSTAT-MCQ8-016 to APSTAT-MCQ8-018 refer to the following stimulus.

Four maintenance tasks, labeled A, B, C, and D, are placed in a uniformly random order. All 24 possible task orders are equally likely.

APSTAT-MCQ8-016

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Four tasks A, B, C, and D are placed in a uniformly random order. What is the probability that task A is first?

- A) $\frac{1}{4}$
- B) $\frac{1}{3}$
- C) $\frac{1}{2}$

D) $\frac{3}{4}$

APSTAT-MCQ8-017

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Four tasks A, B, C, and D are placed in a uniformly random order. What is the probability that tasks A and B are adjacent, in either order?

A) $\frac{1}{4}$

B) $\frac{1}{3}$

C) $\frac{1}{2}$

D) $\frac{2}{3}$

APSTAT-MCQ8-018

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Four tasks A, B, C, and D are placed in a uniformly random order. Given that A is first, why is the conditional probability that B is second equal to $\frac{1}{3}$?

A) Three of the four tasks are equally likely to be first.

B) There are three possible positions for A after the first position is filled.

C) The events "A first" and "B second" are mutually exclusive.

D) After A is fixed first, each of B, C, and D is equally likely to occupy the second position.

APSTAT-MCQ8-019

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A city wants to sample pedestrians using a large downtown area over a full week. Which plan best spreads selection across both place and time?

A) Interview people leaving the busiest station during Friday rush hour.

- B) Randomly select intersection-time blocks across the week, then use a fixed rule to sample pedestrians within each selected block.
- C) Ask volunteers at several intersections to complete an online form later.
- D) Follow one randomly chosen pedestrian and interview everyone that person meets.

APSTAT-MCQ8-020

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a two-proportion test, the observed difference is $\hat{p}_1 - \hat{p}_2 = 0.12$ and the null standard error is 0.04. What is the z statistic?

- A) 3.00
- B) 0.48
- C) 0.33
- D) 0.08

APSTAT-MCQ8-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Over twelve months, both cumulative advertising expenditure and cumulative sales have a very high positive correlation. Why should this correlation be interpreted cautiously?

- A) Correlation cannot be computed for variables measured in dollars.
- B) A positive correlation proves advertising has no effect on sales.
- C) Cumulative variables must always have correlation 0.
- D) Both cumulative totals tend to rise with time, so a shared time trend can create a strong association without establishing a causal effect.

APSTAT-MCQ8-022

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A confidence interval for $\mu_1 - \mu_2$ is $(-3, 5)$. If 7 is added to every observation in population 1's sample and nothing is changed in population 2's sample, what is the corresponding interval after the transformation?

- A) $(-10, -2)$
- B) $(-3, 12)$
- C) $(4, 12)$
- D) $(-21, 35)$

APSTAT-MCQ8-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A packaging manager wants to compare the consistency of fill weights from two machines, not their typical fill levels. Which comparison most directly addresses the question?

- A) Compare only the two sample means.
- B) Compare measures of spread, such as the sample standard deviations or IQRs, for the two machines.
- C) Compare the names and ages of the machines.
- D) Combine all observations and calculate one overall median.

APSTAT-MCQ8-024

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Three children independently choose one of five equally likely colors. What is the probability that at least two children choose the same color?

- A) $13/25$
- B) $12/25$
- C) $3/25$
- D) $1/5$

APSTAT-MCQ8-025

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A test fails to reject H_0 , and the study had power only 0.25 at a scientifically meaningful alternative value. Which conclusion is most appropriate?

- A) H_0 has a 0.75 probability of being true.
- B) The meaningful alternative has been ruled out.
- C) The result is inconclusive because a study with low power often fails to detect a meaningful departure from H_0 .
- D) The Type I error probability must be 0.75.

APSTAT-MCQ8-026

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A frequency table has values 0, 1, 2, and 3 with frequencies 2, 5, 6, and 2, respectively. What is the median of the 15 observations?

- A) 0
- B) 1
- C) 1.5
- D) 2

APSTAT-MCQ8-027

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

Researchers want to compare mean sound levels in two neighborhoods. Which plan best controls measurement differences while supporting population inference?

- A) Randomly select comparable location-time combinations in each neighborhood and use the same calibrated meter and protocol for every measurement.
- B) Measure the loudest location in neighborhood 1 at night and the quietest location in neighborhood 2 in the morning.
- C) Let residents submit sound readings from any phone app they choose.
- D) Use one meter in neighborhood 1 and a differently calibrated meter in neighborhood 2.

APSTAT-MCQ8-028

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Before collecting data, a researcher expects a population proportion near 0.04. What minimum sample size is needed so that the expected numbers of successes and failures are both at least 10?

- A) 240
- B) 250
- C) 260
- D) 400

APSTAT-MCQ8-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

In a one-sample t-test, $\bar{x}$, μ_0 , and n are held fixed while the sample standard deviation doubles. What happens to the t statistic?

- A) Its magnitude doubles.
- B) It increases by 2.
- C) Its magnitude is cut in half.
- D) It is unchanged.

APSTAT-MCQ8-030

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A researcher selects clusters of students and computes a confidence interval for a proportion as if all students were independently sampled. Students within the same cluster tend to respond similarly. Which assessment is best?

- A) The interval is automatically valid because the total number of students is large.
- B) Clustering makes the sample proportion equal the population proportion.
- C) The interval must be too wide because similar responses increase information.

- D) Ignoring within-cluster dependence can understate the standard error and make the interval too narrow.

APSTAT-MCQ8-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

A simulation will estimate the probability that a sequence of 12 fair coin flips contains a run of at least 3 consecutive heads. What should be recorded from each simulated sequence?

- A) Whether the sequence contains at least one run of 3 or more consecutive heads
- B) The total number of heads, regardless of their order
- C) Only the result of the twelfth flip
- D) The number of heads among the first three positions only

APSTAT-MCQ8-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a random sample of 400 observations, $\hat{p} = 0.62$. Using $z^* = 1.96$, which is the approximate 95% confidence interval for p ?

- A) (0.596, 0.644)
- B) (0.582, 0.658)
- C) (0.572, 0.668)
- D) (0.522, 0.718)

APSTAT-MCQ8-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Researchers plan an observational study of screen use and later sleep duration among adolescents. Which plan is strongest for measuring an association while preserving temporal order?

- A) Ask volunteers once whether they think screens harm sleep.

- B) Select a probability sample, measure baseline screen use, then follow participants and record later sleep duration using a common protocol.
- C) Randomly assign adolescents to report either screen use or sleep, but not both.
- D) Select only adolescents already attending a sleep clinic.

APSTAT-MCQ8-034

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For an independent two-sample mean procedure, $SE = \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$. If n_1 , s_1 , and s_2 are held fixed while n_2 becomes extremely large, what happens to the standard error?

- A) It increases without bound.
- B) It approaches $s_2/\sqrt{n_2}$.
- C) It becomes exactly zero.
- D) It approaches $s_1/\sqrt{n_1}$ because uncertainty from sample 1 remains.

Questions APSTAT-MCQ8-035 to APSTAT-MCQ8-037 refer to the following stimulus.

A store records x = hours since opening and y = customer wait time. Express-service customers, who mostly arrive later, form a band near 5 minutes; standard-service customers, who mostly arrive earlier, form a band near 20 minutes. Within each service type, the scatterplot shows little relationship between x and y . When both types are combined, the overall scatterplot slopes downward.

APSTAT-MCQ8-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Express customers mostly arrive later with waits near 5 minutes, while standard customers mostly arrive earlier with waits near 20 minutes. What direction should the combined correlation between hours since opening and wait time have?

- A) Negative
- B) Positive
- C) Exactly zero
- D) It cannot have a sign because service type is categorical.

APSTAT-MCQ8-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Within each service type, the scatterplot of hours since opening and wait time shows little relationship. Which description is most accurate?

- A) Both within-type correlations must be strongly negative.
- B) Each within-type correlation is likely near zero even though the combined correlation is negative.
- C) The combined correlation must equal the average of the two within-type correlations.
- D) Correlation cannot be computed within a service type.

APSTAT-MCQ8-037

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

The combined scatterplot shows a negative association between hours since opening and wait time, but each service type separately shows little association. Which conclusion is best?

- A) Wait time necessarily falls as the day progresses for both service types.
- B) Hours since opening causes customers to switch service types.
- C) Service type is associated with both arrival time and wait time, so the combined negative association should not be interpreted as a common within-type trend.
- D) A regression line fitted to all customers eliminates the need to examine groups.

APSTAT-MCQ8-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

A continuous probability density has height 1.5 over part of its support. Which statement is correct?

- A) A density height can exceed 1 because probabilities are areas under the curve, and the total area must equal 1.
- B) The model is invalid because every density height must be at most 1.
- C) The probability of any exact value in that region is 1.5.
- D) The total area under the density must equal its maximum height.

APSTAT-MCQ8-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A 95% confidence interval for $p_1 - p_2$ is $(-0.01, 0.01)$. Before seeing the data, differences within ± 0.03 were defined as practically negligible. Which interpretation is best?

- A) The interval proves $p_1 = p_2$ exactly.
- B) The interval shows a statistically significant nonzero difference.
- C) The practical threshold must be reduced to ± 0.01 after seeing the data.
- D) Although exact equality is not proved, all values in the interval are within the prespecified range of practical similarity.

APSTAT-MCQ8-040

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A bar chart compares proportions 0.48 and 0.52 but uses a vertical axis from 0.45 to 0.55. Which assessment is most appropriate?

- A) The actual difference is 40 percentage points because the bars fill 40% of the displayed axis.
- B) The actual difference is 4 percentage points, and the truncated axis can visually exaggerate it.
- C) The two proportions are equal because both are near 0.50.
- D) A bar chart must always begin at the smaller observed proportion.

APSTAT-MCQ8-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Researchers classify animal behavior from video and want to assess measurement reliability. Which plan is best?

- A) Have one rater recode only the clips that were difficult.
- B) Tell a second rater the first rater's classifications before recoding.

- C) Randomly select a subset of clips and have two trained raters independently classify them while blinded to each other's decisions.
- D) Use two raters on different species and compare their overall percentages.

APSTAT-MCQ8-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Nine positive lengths have median 4 centimeters. Each length is transformed to the area of a square with that side length, $y = x^2$. What is the median of the nine transformed values?

- A) 2 square centimeters
- B) 4 square centimeters
- C) 8 square centimeters
- D) 16 square centimeters

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ8-001, Correct: C, Correct Answer: 1.50, Explanation: Apply the supplied tail-sum identity: $E(X) = 0.80 + 0.50 + 0.20 = 1.50$.

APSTAT-MCQ8-002, Correct: A, Correct Answer: The display suggests right skew, although a boxplot does not show every feature of the distribution., Explanation: Greater spread above the median and a longer upper whisker are consistent with a longer right tail. A boxplot gives limited shape information, so it suggests rather than proves a particular distributional form.

APSTAT-MCQ8-003, Correct: D, Correct Answer: Test 2 has the smaller p-value because the t distribution has lighter tails at the larger degrees of freedom., Explanation: With the same $|t|$, a t distribution with more degrees of freedom places less probability in the two tails beyond that magnitude. Its p-value is therefore smaller, though not zero.

APSTAT-MCQ8-004, Correct: B, Correct Answer: $H_0: p_{\text{auto}} = p_{\text{manual}}$; $H_a: p_{\text{auto}} > p_{\text{manual}}$, Explanation: Hypotheses use population proportions, equality is placed in H_0 , and "higher" determines a right-tailed alternative for $p_{\text{auto}} - p_{\text{manual}}$.

APSTAT-MCQ8-005, Correct: C, Correct Answer: 6, Explanation: Substituting the fitted point into $\hat{y} = b_0 + 2.5x$ gives $26 = b_0 + 2.5(8)$. Therefore $b_0 = 26 - 20 = 6$.

APSTAT-MCQ8-006, Correct: D, Correct Answer: Have assessors who do not know treatment assignments evaluate the final mobility recordings., Explanation: Blinding outcome assessors prevents knowledge of treatment from systematically influencing mobility scores, even when participants and treatment providers cannot be blinded.

APSTAT-MCQ8-007, Correct: A, Correct Answer: Randomly select many batches and inspect one randomly selected item from each selected batch., Explanation: Spreading observations across randomly selected batches and using one item per batch reduces within-batch dependence while giving batches and items a defined selection mechanism.

APSTAT-MCQ8-008, Correct: B, Correct Answer: 544, Explanation: Solve $0.020 = \sqrt{[0.32(0.68)/n]}$. Thus $n = 0.32(0.68)/(0.020^2) = 0.2176/0.0004 = 544$.

APSTAT-MCQ8-009, Correct: D, Correct Answer: Dataset A, because 2 meters equals 200 centimeters, Explanation: Standard deviations must be expressed in common units before comparison. Dataset A's standard deviation is 200 centimeters, which exceeds 150 centimeters.

APSTAT-MCQ8-010, Correct: C, Correct Answer: The design supports a causal conclusion for, and generalization to, the company's employee population represented by the sampling frame., Explanation: Random assignment supports causation, and the simple random sample supports generalization to the employee population covered by the sampling frame.

APSTAT-MCQ8-011, Correct: A, Correct Answer: For each participant, use assigned exercise duration as x and after-minus-before heart-rate change as y ., Explanation: The quantitative explanatory value and quantitative response change must remain paired for each participant. A regression can then describe how mean heart-rate change varies with assigned duration.

APSTAT-MCQ8-012, Correct: B, Correct Answer: 0.60, Explanation: Because A and B are mutually exclusive, $P(A \mid A \text{ or } B) = P(A)/[P(A) + P(B)] = 0.30/0.50 = 0.60$.

APSTAT-MCQ8-013, Correct: C, Correct Answer: Randomly select combinations from a complete list of products, store locations, and eligible time blocks, then record stock status., Explanation: The target observational unit is an item-location-time combination. Randomly sampling from a complete list gives each such combination a defined chance of selection.

APSTAT-MCQ8-014, Correct: D, Correct Answer: 36, Explanation: Adding 10 does not affect variance, while multiplying by -3 multiplies variance by $(-3)^2 = 9$. Thus $s_y^2 = 9(4) = 36$.

APSTAT-MCQ8-015, Correct: B, Correct Answer: A one-sample t-test of $H_0: \mu_{\text{error}} = 0$ against $H_a: \mu_{\text{error}} \neq 0$, Explanation: Each day yields one quantitative forecast error. Unbiasedness on average corresponds to a population mean error of zero, so a two-sided one-sample t-test is appropriate.

APSTAT-MCQ8-016, Correct: A, Correct Answer: 1/4, Explanation: By symmetry, task A is equally likely to occupy any of the four positions, so $P(A \text{ first}) = 1/4$.

APSTAT-MCQ8-017, Correct: C, Correct Answer: 1/2, Explanation: Treat A and B as one block. The block and the other two tasks can be ordered in $3!$ ways, and A and B can occur in either internal order, giving $2(3!) = 12$ favorable orders among $4! = 24$. Thus the probability is $1/2$.

APSTAT-MCQ8-018, Correct: D, Correct Answer: After A is fixed first, each of B, C, and D is equally likely to occupy the second position., Explanation: Conditioning on A first leaves three tasks randomly ordered in the remaining positions. Each is equally likely to be second, so $P(B \text{ second} \mid A \text{ first}) = 1/3$.

APSTAT-MCQ8-019, Correct: B, Correct Answer: Randomly select intersection-time blocks across the week, then use a fixed rule to sample pedestrians within each selected block., Explanation: Randomly selecting both locations and times spreads coverage across the target setting, and a prespecified within-block rule avoids interviewer convenience selection.

APSTAT-MCQ8-020, Correct: A, Correct Answer: 3.00, Explanation: The statistic is $z = (\text{observed difference} - \text{null difference})/\text{SE}_0 = (0.12 - 0)/0.04 = 3.00$.

APSTAT-MCQ8-021, Correct: D, Correct Answer: Both cumulative totals tend to rise with time, so a shared time trend can create a strong association without establishing a causal effect., Explanation: A common upward time trend can produce a high correlation between cumulative variables even when the month-to-month relationship does not support a causal claim.

APSTAT-MCQ8-022, Correct: C, Correct Answer: (4, 12), Explanation: Adding 7 to every observation in group 1 increases $\bar{x}_1 - \bar{x}_2$ and both interval endpoints by 7 while leaving the standard error unchanged. The new interval is $(-3 + 7, 5 + 7) = (4, 12)$.

APSTAT-MCQ8-023, Correct: B, Correct Answer: Compare measures of spread, such as the sample standard deviations or IQRs, for the two machines., Explanation: Consistency concerns variability rather than center. Standard deviation and IQR measure spread, whereas means compare typical levels.

APSTAT-MCQ8-024, Correct: A, Correct Answer: 13/25, Explanation: The complement is that all three choices differ. After any first color, the next two must choose different unused colors, giving probability $(4/5)(3/5) = 12/25$. Therefore the requested probability is $1 - 12/25 = 13/25$.

APSTAT-MCQ8-025, Correct: C, Correct Answer: The result is inconclusive because a study with low power often fails to detect a meaningful departure from H_0 ., Explanation: Power 0.25 means the test rejects only about 25% of the time at that meaningful alternative. A failure to reject from such a study provides weak evidence against that alternative.

APSTAT-MCQ8-026, Correct: D, Correct Answer: 2, Explanation: With 15 observations, the median is the eighth ordered value. The first 2 values are 0 and the next 5 are 1, so positions 8 through 13 are 2; therefore the median is 2.

APSTAT-MCQ8-027, Correct: A, Correct Answer: Randomly select comparable location-time combinations in each neighborhood and use the same calibrated meter and protocol for every measurement., Explanation: Random location-time selection supports population inference, and one calibrated instrument and protocol prevent measurement method from being confounded with neighborhood.

APSTAT-MCQ8-028, Correct: B, Correct Answer: 250, Explanation: The expected success count requires $0.04n \geq 10$, so $n \geq 250$. At $n = 250$, the expected failure count is $0.96(250) = 240$, which also exceeds 10.

APSTAT-MCQ8-029, Correct: C, Correct Answer: Its magnitude is cut in half., Explanation: The statistic is $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$. Doubling s doubles the denominator while leaving the numerator fixed, so the magnitude of t is halved.

APSTAT-MCQ8-030, Correct: D, Correct Answer: Ignoring within-cluster dependence can understate the standard error and make the interval too narrow., Explanation: Similar responses within clusters provide less independent information than the same number of independently sampled students. An independence-based standard error can therefore be too small.

APSTAT-MCQ8-031, Correct: A, Correct Answer: Whether the sequence contains at least one run of 3 or more consecutive heads, Explanation: Each simulated sequence should be classified according to the event of interest. The estimated probability is the proportion of sequences whose longest head run is at least 3.

APSTAT-MCQ8-032, Correct: C, Correct Answer: (0.572, 0.668), Explanation: $SE = \sqrt{[0.62(0.38)/400]} \approx 0.0243$, so the margin is $1.96(0.0243) \approx 0.0476$. The interval is 0.62 ± 0.0476 , or approximately (0.572, 0.668).

APSTAT-MCQ8-033, Correct: B, Correct Answer: Select a probability sample, measure baseline screen use, then follow participants and record later sleep duration using a common protocol., Explanation: A probability sample supports generalization, measuring screen use before later sleep preserves temporal order, and a common protocol improves measurement consistency. The observational design still does not by itself establish causation.

APSTAT-MCQ8-034, Correct: D, Correct Answer: It approaches $s_1/\sqrt{n_1}$ because uncertainty from sample 1 remains., Explanation: As n_2 grows, s_2^2/n_2 approaches zero, but the fixed contribution s_1^2/n_1 remains. Therefore the total standard error approaches $\sqrt{(s_1^2/n_1)} = s_1/\sqrt{n_1}$ rather than zero.

APSTAT-MCQ8-035, Correct: A, Correct Answer: Negative, Explanation: Later observations are mostly express customers with low waits, while earlier observations are mostly standard customers with high waits. This arrangement produces a downward overall association and negative correlation.

APSTAT-MCQ8-036, Correct: B, Correct Answer: Each within-type correlation is likely near zero even though the combined correlation is negative., Explanation: The stimulus states that wait time has little relationship with time inside either service band. The overall negative pattern is produced by the groups occupying different regions of the plot.

APSTAT-MCQ8-037, Correct: C, Correct Answer: Service type is associated with both arrival time and wait time, so the combined negative association should not be interpreted as a common within-type trend., Explanation: Service type acts as a grouping variable connected to both x and y. Aggregating the groups creates an association that is absent within each group.

APSTAT-MCQ8-038, Correct: A, Correct Answer: A density height can exceed 1 because probabilities are areas under the curve, and the total area must equal 1., Explanation: Density is probability per unit on the horizontal scale, not point probability. A curve may exceed height 1 over a sufficiently narrow interval while retaining total area 1.

APSTAT-MCQ8-039, Correct: D, Correct Answer: Although exact equality is not proved, all values in the interval are within the prespecified range of practical similarity., Explanation: The interval includes zero, so it does not establish a nonzero difference, but its entire range lies inside the predeclared negligible band from -0.03 to 0.03 . This supports practical similarity without proving exact equality.

APSTAT-MCQ8-040, Correct: B, Correct Answer: The actual difference is 4 percentage points, and the truncated axis can visually exaggerate it., Explanation: The numerical difference is $0.52 - 0.48 = 0.04$, or 4 percentage points. Beginning the axis at 0.45 magnifies the visual height difference.

APSTAT-MCQ8-041, Correct: C, Correct Answer: Randomly select a subset of clips and have two trained raters independently classify them while blinded to each other's decisions., Explanation: Independent blinded ratings of the same randomly selected clips permit direct assessment of agreement without one rater influencing the other.

APSTAT-MCQ8-042, Correct: D, Correct Answer: 16 square centimeters, Explanation: Squaring is strictly increasing for positive values, so it preserves their order. The original middle value 4 therefore remains the middle observation after transformation and becomes $4^2 = 16$ square centimeters.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ8

· APSTAT-MCQ8-001 to 005: Unit 2 a bounded nonnegative count uses a supplied tail-sum identity to calculate $E(X)$ from three survival probabilities; Unit 1 a boxplot with a low-positioned median and long upper half cautiously suggests right skew; Unit 4 equal $|t| = 2.1$ at $df = 5$ and $df = 50$ compares tail areas and finds the larger- df p-value smaller; Unit 3 automated-versus-manual on-time proportions formulate a right-tailed population hypothesis; Unit 5 equipment load and energy use recover the regression intercept from one fitted point and slope. Misconceptions include averaging tail probabilities, claiming a boxplot proves exact shape, treating equal t statistics as equal evidence across df , using sample statistics in hypotheses, and dividing rather than subtracting the slope contribution.

· APSTAT-MCQ8-006 to 010: Unit 1 visible physical-therapy programs blind video outcome assessors; Unit 2 an inspection study samples one item from each randomly selected production batch to limit within-batch dependence; Unit 3 reverses the confidence-interval standard-error formula to recover $n = 544$ from $\hat{p} = 0.32$ and $SE = 0.020$; Unit 1 compares standard deviations after converting 2 meters to 200 centimeters; Unit 4 an employee SRS followed by random assignment supports both company-population generalization and a causal conclusion. Misconceptions include assuming blinding is all-or-none, treating many same-batch items as independent, failing to square SE, comparing numerical spreads in different units, and confusing random selection with random assignment.

· APSTAT-MCQ8-011 to 015: Unit 5 assigned exercise duration is paired with each participant's after-minus-before heart-rate change for regression; Unit 2 mutually exclusive alarm causes use $P(A | A \text{ or } B) = 0.30/0.50$; Unit 3 out-of-stock prevalence samples item-location-time combinations from a complete frame; Unit 1 the transformation $y = -3x + 10$ multiplies sample variance 4 by 9; Unit 4 daily actual-minus-forecast temperature errors use a two-sided one-sample mean test against zero. Misconceptions include replacing participant-level pairs with aggregate means, using an unconditional probability after conditioning, sampling only reported stockouts, multiplying variance by -3 , and choosing a two-group or proportion procedure for one quantitative error variable.

· APSTAT-MCQ8-016 to 020: Unit 2 a random ordering of four maintenance tasks finds $P(A \text{ first})$, counts A-B adjacency by a two-order block, and interprets $P(B \text{ second} | A \text{ first})$ from the three remaining equally likely tasks; Unit 1 downtown pedestrian sampling selects random intersection-time blocks and applies a fixed within-block rule; Unit 3 divides an observed two-proportion difference 0.12 by null SE 0.04. Misconceptions include counting positions unequally, omitting one internal block order, retaining four possibilities after conditioning, sampling only one busy period, and multiplying rather than standardizing a difference.

· APSTAT-MCQ8-021 to 025: Unit 5 cumulative advertising and sales totals warn that a shared time trend can create high correlation without causation; Unit 4 adding 7 only to group 1 shifts a $\mu_1 - \mu_2$ interval from $(-3, 5)$ to $(4, 12)$; Unit 1 machine consistency calls for comparing spreads rather than centers; Unit 2 three independent choices among five colors use the all-distinct complement to find a

match probability 13/25; Unit 3 a failure to reject from a study with power 0.25 at a meaningful alternative is judged inconclusive. Misconceptions include causal interpretation of trending cumulative variables, shifting only one interval endpoint, equating consistency with mean, counting only one matching pattern, and treating low-power non-significance as evidence of no effect.

- APSTAT-MCQ8-026 to 030: Unit 1 a four-value frequency table locates the eighth of 15 observations to obtain median 2; Unit 4 neighborhood sound levels use random location-time samples and one calibrated meter; Unit 3 anticipated $p = 0.04$ reverses the expected-success condition to $n = 250$; Unit 4 doubling s while fixing $\bar{x}$, μ_0 , and n halves $|t|$; Unit 3 a cluster-sampled proportion warns that ignored within-cluster dependence understates SE and interval width. Misconceptions include averaging category labels, confounding neighborhood with instrument, checking failures but not rare successes, adding 2 to t , and treating clustered students as independent.

- APSTAT-MCQ8-031 to 035: Unit 2 a 12-flip simulation records whether each sequence contains a head run of length at least 3; Unit 3 $\hat{p} = 0.62$ and $n = 400$ construct a 95% interval (0.572, 0.668); Unit 1 a prospective probability-sample cohort measures screen use before later sleep with a common protocol; Unit 4 as n_2 grows with sample 1 fixed, the independent two-mean standard error approaches $s_1/\sqrt{n_1}$ rather than zero; Unit 5 a grouped wait-time stimulus identifies a negative combined correlation from later low-wait express customers and earlier high-wait standard customers. Misconceptions include recording only total heads instead of runs, omitting z^* or $\sqrt{n}$, inferring causation from temporal order alone, assuming one enormous group removes uncertainty from the other, and claiming a categorical grouping prevents an overall correlation.

- APSTAT-MCQ8-036 to 040: Unit 5 the wait-time stimulus recognizes near-zero within-service correlations and identifies service type as the source of the aggregated downward association; Unit 2 explains how a continuous density height can exceed 1 while total area remains 1; Unit 3 an interval $(-0.01, 0.01)$ lies entirely inside a prespecified ± 0.03 practical-similarity band without proving equality; Unit 1 a truncated bar-chart axis exaggerates an actual four-percentage-point difference. Misconceptions include averaging group correlations to obtain the overall value, equating density height with point probability, treating interval inclusion of zero as proof of exact equality, and measuring differences relative to the displayed axis span.

- APSTAT-MCQ8-041 to 042: Unit 1 independent blinded double-coding of a random video subset assesses behavioral-classification reliability; Unit 1 a strictly increasing square transformation of nine positive lengths preserves order and maps median 4 to median area 16. Misconceptions include allowing one rater to influence another or using different clips, and squaring the number of observations rather than the middle value.

- Unit counts for APSTAT-MCQ8: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: uniformly random ordering of four maintenance tasks, APSTAT-MCQ8-016 to 018; grouped express-versus-standard customer wait-time scatterplot, APSTAT-MCQ8-035 to 037.